A Project Report on

“LOW COST ‘PAPER BRICK’ FOR FINANACIALLY MARGINALIZED SECTION OF URBAN POOR”

Submitted in partial fulfillment of the requirement for the award of the Degree of

BACHELOR OF TECHNOLOGY

To

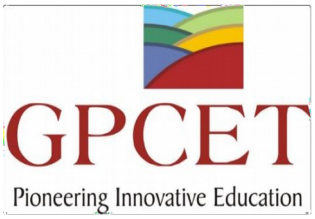
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DEPARTMENT OF CIVIL ENGINEERING

G. PULLAIAH COLLEGE OF ENGINEERING AND TECHNOLOGY

(Autonomous)

(Accredited by NAAC of UGC with “A” Grade, Accredited by NBA (ECE, CSE & EEE) Approved by AICTE, New Delhi, Recognized by UGC under 2 (f) & 12 (B) & Permanently Affiliated to Jawaharlal Nehru Technological University Anantapur)
Pasupula (v), Nandikotkur Road, Kurnool – 518 452, Andhra Pradesh

2019-2023

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DEPARTMENT OF CIVIL ENGINEERING

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CERTIFICATE

This is to certify that the project report entitled **"LOW COST 'PAPER BRICK' FOR FINANCIALLY MARGINALIZED SECTION OF URBAN POOR"** being submitted by **CHAKALI BHARATH (19AT1A0109) , KADU ABDUL JABBAR (19AT1A0101), SHAK ABDUL MUBASHEER (19AT1A0102) , BOMMA AKHIL (19AT1A0104), KALUBURIGI MALLIKARJUNA (19AT1A0124) , KUMMARI SRI HARI (19AT1A0145) , BANDARI SUNIL KUMAR (19AT1A0150)** in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Civil Engineering of Jawaharlal Nehru Technological University Anantapur, Ananthapuramu is a record of bonafide work carried out by them under my guidance and supervision.

The results embodied in this project report have not been submitted to any other university or institute for the award of any Degree or Diploma.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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“LOW COST ‘PAPER BRICK’ FOR FINANCIALLY MARGINALIZED SECTION OF URBAN POOR”

ABSTRACT:

Civil engineering is one of the oldest and broadest engineering professions. It focuses on the infrastructure necessary to support a civilized society. Civil engineering is a professional engineering discipline that deals with the design, construction, and maintenance of the physical and naturally built environment. Including public works such as roads, bridges, canals, dams, airports.

In this age of sustainability, urbanization and urban rich-poor division, it has been inevitable to investigate and find the alternative methods and affordable options for dwelling materials for the financially marginalized section of population in an urban environment.

Among the widely used materials in building dwelling for the urban poor, brick is one of the major component used for construction and it has a significance impact on the construction costs of urban dwelling. In this aspect, a study has been conducted and the alternatives were worked out with abundantly available materials in urban pockets such as paper granules, Portland cement, and PET (Polyethylene Terephthalate) bottles

The study concludes that the construction cast of urban poor dwelling can be reduced drastically with the usage of cost effective alternative for conventional brick, which makes up the major part of the construction components.

KEYWORDS :

Cost Effective Paper brick, PET bottles, sustainable building materials, paper as building materials, shelters for urban poor, low cost housing.

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CHAPTER - 1
INTRODUCTION

A brick is a form of building substance used to create walls, sidewalls, and other masonry components. The word brick refers to a unit made mainly of clay, but it is now also used colloquially to refer to units made of other materials or other chemically cured building pieces. Bricks can be linked together with cement, glue, or by interlocking. Bricks are typically made in bulk amounts at brickworks in a variety of classes, types, materials, and sizes that differ by area.

As we know that the bricks are widely used as constructional material around the world and demand for bricks is continuously rising due to increase in population, it is challenged to convert the industrial wastes to useful building and construction materials, this study investigates the potential use of waste paper, quarry dust, GGBS(Ground Granulated Blast-Furnance Slag) for producing a low-cost and light weight composite brick as a building material. A rectangular construction element made of concrete or clay is referred to as a block by a similar word, but a block is typically bigger than a brick.

Aggregates of enlarged clay are used to create lightweight pavers, also known as lightweight blocks. Fired bricks, also known as artificial stone, have been used as construction elements since about 4000 BC. They are among the toughest and longest-lasting ones. A mechanical binder, such as straw, is an extra component of air-dried bricks, also referred to as "mud bricks," which have a longer past than burned bricks. Bricks are placed in courses and a variety of designs known as bonds, which are all jointly referred to as brickwork. Various types of mortar may be used to hold the bricks together to create a sturdy structure.

One of the first made building materials in history is the brick. At least 14,000 years ago, in Egypt, hand-formed, sun-dried clay panels were found. The majority of the structures in ancient Mesopotamia were made of clay bricks at the time, and clay was also the most important raw material. Bricks were used for the first time in history when they were used to build the ancient metropolis of Ur (present-day Iraq) around 8,000 BC. Since 5,000 BC, people have known how to preserve ceramic panels through firing. Due to the use of both fired and unfired bricks by civilizations along the Euphrates, Tigris, and Indus rivers, fired bricks underwent further refinement. The Romans used hot stones, and they later brought them to England and began using them there.

{ FACES OF BRICKS }



Bricks used in the creation of houses, buildings, and other structures are usually classified into two kinds. Clay brick made from clay and baked at a high temperature, as well as sand brick or mortar brick. The clay brick is crimson, while the sand brick is whitish. Brick is an essential component in construction and is commonly used as a block in high-cost buildings. When compared to sand bricks, clay bricks have a higher tensile strength. The mortar is a combination of sand, cement, and water that is used to make sand blocks. Sand brick is simpler, less expensive, requires less upkeep, and is more commonly manufactured.

Furthermore, traditional brick building materials are reliant on natural resources that are becoming rare. The process of extracting materials from the hills to create cement causes a slew of natural issues and puts people at risk of landslides. Furthermore, extensive cement and brick production sites will cause air pollution, ecosystem problems, and severe noise pollution. Furthermore, the use of cement or portland cement is promoting carbon emissions and increasing the impact on the environment. In order to overcome these problems, there is a need to provide viable solutions by reducing the use of natural resources and reusing waste material. Many scholars have attempted to discover an appropriate substitute substance for cement and sand in the manufacture of bricks. Using cotton wastes and limestone powder refuse as a fine aggregate in brick has enabled for a more cost-effective and lighter alternative that may offer substantial labor and shipping savings.

The wastes of limestone powder and timber saw dust are mixed to create a lightweight composite for use as construction substance. The characteristics of hardened concrete blocks with expanded polystyrene beads as sand substitute are examined in this research. The findings of the tests show that waste limestone sawdust, both with and without waste glass powder, can be used to make masonry blocks. with suitable mechanical & physical.

"Papercrete" is one of such materials attracting public interest. papercrete is a complex material comprising of portland cement, waste paper, water and / or sand. the combination of these materials, which may provide a way to provide affordable housing on a large scale. papercrete have been reported: to be cheap alternative building material: to have good sound absorption and thermal insulation: to be a light weighted and fire- resistant material. Although it was invented over 80 years ago, it has only recently been rediscovered, and only a few studies have been conducted to determine its structural suitability. It should be noted that papercrete is a comparatively new concept with limited scope. fibrous concrete is a mixture of paper, Portland cement, and water. there is no production of by products or excessive energy.

Papercrete is known by alternative names such as fibrous concrete, padobe and fidobe. fibrous concrete is a mixture of paper, portland cement and water. there is no production of by products or excessive energy use in the production of papercrete. padobe has no portland cement. here, instead of portland cement, clay is the binding material. it is a mix of paper, water and earth with clay. papercrete is a recently developed construction material which consists of re-pulped paper fiber with portland cement or clay. it is recognized as an environment-friendly was discovered by eric patterson and mike mccain independently and they named their invention padobe and fibrous cement. Papercrete is a new composite material using waste paper as a partial replacement of Portland cement. By using the waste paper, papercrete reduces the amount of cement use as well as it makes environment friendly building material. Papercrete is an alternative building material which reduces dead load for the main structure. The basic constituents of papercrete are water and any kind of paper (newspaper, cardboard, glossy magazine stock, advertising brochures, junk mail or any other type of paper) are used for manufacturing of paper brick. eric patterson and mike mccain independently came up with papercrete, which they called padobe and fibrous cement. papercrete is a newly developed composite material using waste paper as a partial replacement of Portland cement. by using the waste paper, papercrete reduces the amount of cement use as well as a number of other environmental concerns. "papercrete" is one such material attracting public interest. papercrete is a complex material composed of portland cement, waste paper, water, and/or sand. the combination of these materials, which may provide a way to provide affordable housing on a large scale.

And / or sand. the combination of these materials, which may provide a way to provide affordable housing on a large scale. papercrete have been reported: to be cheap alternative building material: to have good sound absorption and thermal insulation: to be a light weighted and fire- resistant material.



PAPER BRICK (Made With Papercrete)



PAPERCRETE

(Mixture of Paper, Cement, PET Bottles, Water (to be added))

A brick is material of building utilized to make partition walls, road pavements and other elements in masonry construction. Traditionally, the term brick relate to a unit tranquil of clay, but it is now used to refer any rectangular units laid in mortar. A brick can be combination of claybearing soil, sand, and lime, or concrete materials. Bricks are made in many classes, sort, materials, and sizes which differ with region and time period, and are produced in mass quantities. Block is a analogous term referring to a rectangular brick.

ENVIRONMENTAL SUATAINABILITY

The environment impact of paper is significant, which has led to changes in industry. With the use of modern technology, harvesting of wood, disposable paper has become a cheap commodity which has led to a high level of consumption and waste. The production and use of paper has a number of adverse effects on the environment which are known as paper pollution. Discarded paper is a major component of it. Taking this issue into account, construction material known as papercrete is invented. Papercrete is an innovative composite material developed to build an environmental friendly house made up of paper, cement and water. It has been reported to be a cheap alternatively building construction material, to have a good absorption and thermal insulation, to be a light weight and fire resistant material. The paper used in making are those that need to be recycled. It is perceived as an environmental friendly material due to the vital recycled content. It is made by mixing the dry ingredients with water to form slurry and cast the slurry into blocks and let it dry. It holds its shape even when it is wet and is remarkably strong. Using papercrete helps to reduce waste gives people job in making papercrete and also help in developing the economic state of each individual. The product is highly sustainable as it was around 80% recycled material, hence capturing carbon dioxide emission by using waste as raw material. Papercrete is an emerging new concept however it has limited scope.

Approximately, the global production of paper is up to 407 million metric tons in 2014, where,47% is from the Asian countries. About 45% of the paper gets wasted and ends up in the trash.Additionally, about 50% of the waste that companies produce consists of paper.Every year paper accounts for 25% of landfill waste and 33% of municipal waste. The present recovery and utilization of waste paper by the paper mills in India is about 3.0 million tons annually.This translates towards a recovery of 27% of the total paper and paperboard consumed, which is too little when compared to the recovery rates in developed countries like Germany (73 %), Sweden (69 %), Japan (60 %),Western Europe (56 %), USA (49 %) and Italy (45 %). Apart from causing deforestation, paper affects the environment by the means of pollution. Paper is the third biggest cause of air pollution in the industrial world. Paper production involves the use of chlorine-based bleaches, Dyes, Varnishes. These toxic materials are hard to degrade i.e., these could not be broken down by the bacteria. This results in contamination of both water and soil. Recycling one ton of paper can save up to 17 trees and about 26,500 liters of water. Hence, it is strongly advocated towards the reuse of waste paper in a safe, environment friendly and cost-effective manner as a building material for the

Urban poor. Most of these lives have extremely limited resources to meet their needs for their survival. Shelter is one of the basic needs, especially in the areas of urban poor. An examination and a study has been conducted to come up with the alternative building materials for urban poor. The materials were expected to be easily available and cost effective to help the environmental sustainability. Moreover, the material must be accessible to meet the standards and the needs of building construction, structural stability and environmental safety. The right to see our full range of tests. We are always happy to talk to you about the suitability of particular tests, so please make contact with the Smithers Pira expert shown if you have questions about the test itself or the support you need for your particular project or issue. As standard, all of our tests are performed in a controlled laboratory environment. However, we have the capability to combine any of the above testing services at a set temperature or after exposure to freezing temperatures or high humidity.

Now a days, a lot of researchers and scholars are moving with the main aim of "environmental sustainability" in their researches. So, that the future scope of their research will be good enough and it will be helpful for global environment. Which leads a better environmental conditions and better human life experience in this earth. So, the main aim of a research should be always to save environment and to sustain the environmental conditions. Furthermore, the researchers will always think their first objective of their project as the environmental sustainability. As our project is about the manufacturing of brick by using ingredients (or) components as cement, paper pulp pieces(wastes), plastic bottle pieces(wastes), water , fly ash. The main motive of our project research is to decrease pollution and land degradation . and it will increase land properties by not making it polluted by the wastes, as we collects the wastes for the preparation of papercrete brick .it is a best alternative for the civil constructions and it will be also helpful in the conservation of environment in a better way. Therefore, our project will be beneficial for the protection of environmental conditions and however it will be better for the human life experience and the main aim and objective of the project is to sustain the environment.

CHAPTER - 2
OBJECTIVES

The objectives of this project [Low Cost 'PAPER BRICK' For Financially Marginalized Section Of Urban Poor] are as follows :

- The main objective of the project is to improve the properties of papercrete brick by utilizing the waste materials like Fly Ash and Lime.
- Recycling paper saves landfill space and reduce the amount of pollution in the air from incineration.
- The main climate-related objective of waste policy is to reduce the greenhouse gas emissions generated by waste, particularly by reducing the methane emissions resulting from treatment at landfills.
- Reducing, reusing, recycling waste helps save landfill space by keeping useful materials out.
- The amount of energy and natural resources needed to produce or collect the raw materials and manufacture the product are reduced.
- Removing such waste from the rest of the waste stream substantially reduces waste volumes for disposal and also allows biodegradable waste to be composted.
- Biodegradable waste can be used for composting or a resource for heat, electricity and fuel by means of incineration or anaerobic digestion.
- These are the three essential components of environmentally-responsible consumer behavior - reduce, reuse and recycle.
- The goal of the 3rs is to educate people on how to prevent excessive and unnecessary waste, and to limit the consumption of non-renewable resources.yjy
- The main objective is to improve the properties of papercrete brick by utilizing the waste materials like fly ash, waste paper pieces, waste plastic bottles.
- To work the properties of papercrete bricks by using the waste materials like fly debris and admixtures.
- The goal of the review is to explore quality principles like hardness, sufficiency and imperviousness to fire.

- Dependence on conventional sources of energy should be reduced and use of non conventional sources of energy should be encouraged.
- Preservation of the generation of resources for the future. The sustainable development goals (SDGs) aim to transform our world.
- They are a call to action to end poverty and inequality, protect the planet, and ensure that all people enjoy health, justice and prosperity.
- These include social progress and equality, environmental protection, conservation of natural resources and stale economic growth. Everybody has the right to a healthy, clean and safe environment.
- This can be achieved by reducing pollution, poverty, poor housing and unemployment.
- To maintain ecological diversity.
- To provide resources for future generations. To maintain employment facilities for people.
- To preserve the diversity of species, Sustainable utilization of species and ecosystem.
- To maintain life-supporting systems and essential ecological processes.
- To find effective alternatives for conventional brick.
- To propose mix design for bricks using material.
- To test the strength and physical properties of bricks with waste material and compare it with conventional bricks.
- A set of papercrete mixes were prepared with cement, waste paper sludge, river sand.
- The major objective of this research program is replacing the costly and scarce conventional building bricks by an innovative and alternative building bricks.
- Lightweight, less water absorption, inflammable, abundant availability, environmental friendly, easily available, cost effective.
- The purpose of present research is to utilize the waste materials like paper and to replace.
- The primary goals are to really use the waste paper to lesson the issue of removal of waste, to demonstrate that the reused paper mash can be a halfway swap for fine total.
- Natural resources as the components for the construction materials.

- It is also suitable for making community rooms, sale booths, storage rooms and dwelling for livestock.
- Papercrete can also be used as a plaster. It can be sprayed on walls to give them good sound and heat insulating properties.
- Helps to save landfill space by keeping useful materials out.
- The dry ingredients are mixed with water in a mixer to form slurry. The slurry is cast into blocks or panels and allowed to dry in the sun.
- Reduces waste volumes for disposal and also allows biodegradable waste t be composted.
- To educate people on how to prevent excessive and unnecessary waste, and to limit the consumption of non-renewable resources.
- The main objective of this project is to optimize the papercrete mix with desirable properties, which satisfies the above mentioned needs. Accumulation of material, experimental procedure, results and discussions, utilization of waste materials.
- To effectively utilize the waste paper to reduce the problem of disposal of waste, to prove that the recycled paper pulp can be a replacement for fine aggregate.
- The objective of the study is to investigate the properties of papercrete bricks that were prepared out of waste paper, cement, pet bottles, water. The following properties were studied and a comparative study with conventional bricks was done.
- Mechanical properties, weight comparison, quality standards like hardness, soundness and fire resistance, environmental compatibility.
- To analyze and improvise the properties of normal papercrete brick by using fly ash and lime.
- To determine the compressive strength of normal papercrete bricks made using fly ash at 14 and 28 days
- To conduct water absorption test and weight test on normal papercrete bricks made using fly ash.
- To compare the compressive strength, water absorption and weight test of papercrete brick with conventional brick.

- To study the compressive strength, absorption test and weight test of papercrete brick manufactured using fly ah, waste paper, waste plastic bottles.
- One of the main objective is to improve "sustainable housing".
- To promote or develop the "cost effective housing techniques".
- To promote or develop the "Eco-Friendly Housing".
- To promote or develop the natural resources for the civil construction.
- The main objective of this research work is t make low cost and eco-friendly bricks to maintain environmental balance, and to reduce the problem of waste paper disposals.
- The goal of the review is to explore the weight examination.
- To use the waste materials like newspaper, fly ash in the process of producing new type of green bricks.
- To construct and study the properties of the papercrete bricks in order to use these material commercially for construction purposes in building.
- The objective of this study is to determine the compressive strength of a light weight brick manufactured using waste paper.
- To produce light weight and fire-resistance solid blocks which are to be used in partition walls of especially high-rise buildings.
- The main objective of our project is "Environmental Sustainability".

CHAPTER - 3
LITERATURE REVIEW

Gaurav Goel and Ajay Kalamdhad (2017) investigated on experimental study of manufacturing eco-friendly lightweight bricks through binary mix of paper mill sludge (PMS) and soil. The mix ratio between PMS and soil vary (0%, 5%, 10%, 15% and 20%) and two firing temperatures at 850°C and 900°C are tested in a kiln. The performance of incorporating PMS, into the mix, tested by evaluating properties such as linear shrinkage, compressive strength, water absorption, mass loss on ignition, and bulk density of bricks as recommended by the relevant Indian and ASTM standard codes. Based on the results, an optimum mix of 10% PMS with both soil types are found suitable for brick production at a firing temperature of 900°C.

Nitin Goyal et al. (2016) investigated a new concept of eco-bricks also known as bottle bricks used as a construction material. These bricks are manufactured by using the non-bio-degradable wastes such as plastic bottles, plastic bags and other non-bio-degradable substances are used to construct garden partition walls and school buildings and houses. By consideration of cost of construction, these bricks are less cost effective and can be a shelter home for the poor. The houses that are constructed with these bricks are bulletproofing and less prone to attack. Hence, the author concluded that the cost and resource efficient building material which can be used in order to deal with the various environmental problems as well as in the reduction of cost of construction.

Rohit Kumar Arya and Rajeev Kansal (2016)this research is to determine the weight, compressive strength, water absorption capacity, fire resistance, hardness etc., of Papercrete brick by using waste papers (newspapers, invitation cards, magazines etc.) in order to determine their aptness for use as a building construction material. While using paper pulp with cement and sand the weight of the brick is approximately 50% lesser than the conventional clay brick. Therefore, papercrete bricks will decrease the dead weight of the structure to a significant amount. Based on the test results, the compression test showed that Papercrete bricks are acceptable for non-load bearing walls only. The water absorption capacity of papercrete brick is more than 20%, which makes it not suitable for water logging and external walls. The total cost of building construction is reduced from 20% to 50% by using these bricks.

Maneeth et al. (2014) in his research an attempt has been made to produce bricks by using waste plastic (PET bottles) in range of 60% to 80% by weight of laterite quarry waste in molten form, which acts as a binder. When this molten plastic was not sufficient for binding capacity, to overcome this problem a little quantity of 60/70 grade bitumen is added in a range of 2% to 5% by weight of soil and this bitumen-plastic resin was mixed with laterite quarry waste to manufacture bricks. Based on the tests, the compressive strength for plastic-soil bricks with 70% plastic content by weight of soil with the binder(bitumen) content of 2% by weight of soil gives a compressive strength of 8.16 N/mm2 which is higher than laterite stone (3.18N/mm2) and has a lesser water absorption(0.9536%) than laterite stone (14.58%).Dr. Chitraet al. (2014) attempted to minimize the presence of pores in conventional concrete by using carbon black powder, which is a waste material from rubber industry as filler. The research reveals that concrete specimen with 2% and 5% carbon black showed good performance with respect to conventional concrete. The author concluded that the addition of carbon black up to 5% as filler is very effective in concrete.

Rajput et al. (2012) researched on the Waste Crete Bricks can be prepared by recycled Paper mills Waste (PW) and cotton waste (CW). Waste Crete Bricks with varying content of cotton waste (1%–5% wt.), Recycle Paper Mills waste (89%–85% wt.) and fixed content of Portland cement (10% wt.) have been prepared and tested as per IS 3495 (Part 1–3): 1992 standards. The brick samples with PW, CW, which have physical and mechanical properties and cement are investigated. By testing observations it is revealed that the bricks with 1%–5% addition of CW and 10% cement to PCW exhibit a compressive strength of 21–23 MPa (with 30% shrinkage) which is several times greater than the conventional clay bricks. The author concludes that the PCW–cement combination can be potentially used in the production of lighter and economical brick material, which can be used as internal partition wall and also concludes that 85% PW–5% CW–10% cement is the optimum composition.

Raut et al.(2011) investigated on the brick prepared with Recycle paper mills waste (RPMW) and cement which is known as waste-create bricks (WCB). RPMW with cement combination and low carbon footprint bricks with varying composition of cement (0%–20% wt) have been prepared and tested as per standards. In this investigation, the author observes that waste-create bricks (WCB) prepared by using RPMW-cement combination is lightweight, shock absorbing and meets compressive strength requirements of ASTM.

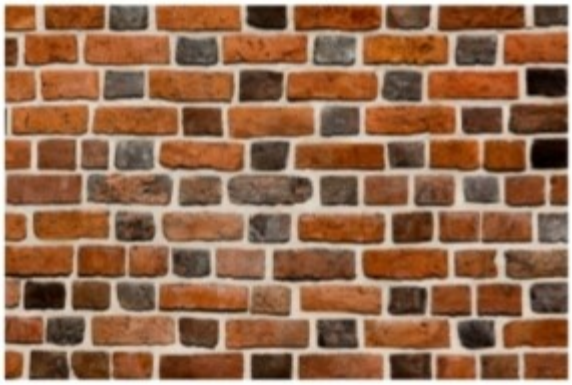
CHAPTER - 4
HISTORY OF BRICKS

A brick is a type of construction material used to build walls, pavements and other elements in masonry construction. Properly, the term brick denotes a unit primarily composed of clay, but is now also used informally to denote units made of other materials or other chemically cured construction blocks. Bricks can be joined using mortar, adhesives or by interlocking. Bricks are usually produced at brickworks in numerous classes, types, materials, and sizes which vary with region, and are produced in bulk quantities.



Block is a similar term referring to a rectangular building unit composed of clay or concrete, but is usually larger than a brick. Lightweight bricks (also called lightweight blocks) are made from expanded clay aggregate. Fired bricks are one of the longest-lasting and strongest building materials, sometimes referred to as artificial stone, and have been used since circa 4000 BC. Air-dried bricks, also known as mud-bricks, have a history older than fired bricks, and have an additional ingredient of a mechanical binder such as straw. Bricks are laid in courses and numerous patterns known as bonds, collectively known as brickwork, and may be laid in various kinds of mortar to hold the bricks together to make a durable structure.

{ A wall constructed in **Flemish bond** with bricks of various shades and lengths. }



Bricks used in the creation of houses, buildings, and other structures are usually classified into two kinds. Clay brick made from clay and baked at a high temperature, as well as sand brick or mortar brick. The clay brick is crimson, while the sand brick is whitish. Brick is an essential component in construction and is commonly used as a block in high-cost buildings. When compared to sand bricks, clay bricks have a higher tensile strength. The mortar is a combination of sand, cement, and water that is used to make sand blocks. Sand brick is simpler, less expensive, requires less upkeep, and is more commonly manufactured.

A brick is material of building utilized to make partition walls, road pavements and other elements in masonry construction. Traditionally, the term brick relate to a unit tranquil of clay, but it is now used to refer any rectangular units laid in mortar. A brick can be combination of clay bearing soil, sand, and lime, or concrete materials.

{ An old brick in **English bond** laid with alternating courses of headers and stretchers. }



The earliest bricks were dried mud-bricks, meaning that they were formed from clay bearing earth or mud and dried (usually in the sun) until they were strong enough for use. The oldest discovered bricks, originally made from shaped mud and dating before 7500 BC, were found at Tell Aswad, in the upper Tigris region and in southeast Anatolia close to Diyarbakir. Mud-brick construction was used at Çatalhöyük, from c. 7,400 BC. Mud-brick structures, dating to c. 7,200 BC have been located in Jericho, Jordan Valley. These structures were made up of the first bricks with dimension 400x150x100 mm.

Methods of Manufacture : Four basic types of brick are un-fired, fired, chemically set bricks, and compressed earth blocks. Each type is manufactured differently for various purposes.

Mud-Brick : Unfired bricks, also known as mud-bricks, are made from a mixture of silt, clay, sand and other earth materials like gravel and stone, combined with tempers and binding agents such as chopped straw, grasses, Fired and unfired brick making process tree

Bark, or dung. Since these bricks are made up of natural materials and only require heat from the Sun to bake, mud bricks have a relatively low embodied energy and carbon footprint.

Shaping Methods : Three main methods are used for shaping the raw materials into bricks to be fired.

1. Molded bricks – These bricks start with raw clay, preferably in a mix with 25– 30% sand to reduce shrinkage. The clay is first ground and mixed with water to the desired consistency. The clay is then pressed into steel moulds with a hydraulic press. The shaped clay is then fired ("burned") at 900–1,000 °C (1,650– 1,830 °F) to achieve strength.

2. Dry-pressed bricks – The dry-press method is similar to the soft-mud moulded method, but starts with a much thicker clay mix, so it forms more accurate, sharper-edged bricks. The greater force in pressing and the longer burn make this method more expensive.

3. Extruded bricks – For extruded bricks the clay is mixed with 10–15% water(stiff extrusion) or 20–25% water (soft extrusion) in a pugmill. This mixture is forced through a die to create a long cable of material of the desired width and depth. This mass is then cut into bricks of the desired length by a wall of wires. Most structural bricks are made by this method as it produces hard, dense bricks, and suitable dies can produce perforations as well. The introduction of such holes reduces the volume of clay needed, and hence the cost. Hollow bricks are lighter and easier to handle, and have different thermal properties from solid bricks. The cut bricks are hardened by drying for 20 to 40 hours at 50 to 150 °C (120 to 300 °F) before being fired. The heat for drying is often waste heat from the kiln.

Various types of Bricks : There are various types of bricks used in masonry.

1. **Common Burnt Clay Bricks :** Clay bricks are fired bricks. These are formed by pressing in moulds or by an extrusion and wire cutting process. Then these bricks are dried and fired in a kiln.
2. **Sand Lime Bricks (calcium silicate bricks) :** These bricks are mixtures of sand and hydrated lime pressed in moulds and cured in a high-pressure steam autoclave.
3. **Concrete Bricks :** Concrete bricks are mixtures of cement, sand and aggregates vibrated in moulds and steam cured.
4. **Fly Ash Clay Bricks :** Fly ash is used along with clay in these bricks. Fly ash is obtained from boilers of thermal power stations.

5. **Fire Clay Bricks :** Fire clay exists at much depth below the surface and is usually mined. Generally, fire clays contain metallic oxides less than surface clays and have more uniform chemical and physical properties.

Recommended Bricks For Construction : Common burnt clay bricks are recommended for construction of a house. The bricks should be burnt in kiln thoroughly so that they become hard and durable. The two bricks should give a hard ringing sound when they are struck together. The bricks should have rectangular shapes with sharp corners and uniform colors. The bricks should be free from cracks, chips and warp ages, large particles of lime and organic matter. First class bricks are suitable for construction of a building.

There are Three Types of Bricks :

1. First class bricks
2. Second class bricks
3. Third class bricks

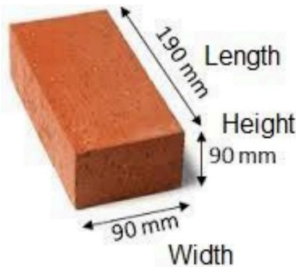
Specifications of First Class Bricks : Made of good earth which is free from saline deposits and are sand molded. Burnt thoroughly without being vitrified and have deep red, cherry and copper color. Regular and uniform in shape and size with sharp and square edges and parallel faces. Must be homogeneous in texture and emit a clear ringing sound on being struct together. free from flaws, cracks, chops, stones and lime. Should not absorb water more than 20% of its own dry weight after 24 hour, immersion in cold water. Have a minimum crushing strength of 105 kg per sq. cm when tested according to the test. should not show appreciable sign of efflorescence either in dry state or subsequent to soaking in water. NOTE : As per guidelines of all wings of he govt, semi-govt depts. and other construction depts., first class brick tiles should be used for construction works.

Specifications of Second Class Bricks : They shall be well burnt or slightly over burnt. They must give clear ringing sound when struct. They may have slight irregularities in size, shape and color. They may have slight chips,flaws or surface crack but must be free from lime or kankar. The minimum crushing strength f second class brick should be 70 kg per sq.cm.

Specifications of Third Class Bricks : These are slightly under burnt or over burnt. They are not uniform in shape, size, and edges

They shall not observe water more than 25% of their own dry weight after 24 hours, immersion in cold water. They have some signs of efflorescence.

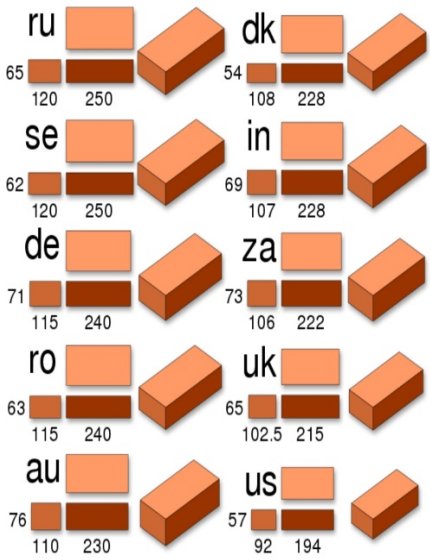
{ Standard Dimension of Brick in **"INDIA" }**



Optimal Dimensions, Characteristics, And Strength : For efficient handling and laying, bricks must be small enough and light enough to be picked up by the bricklayer using one hand (leaving the other hand free for the trowel). Bricks are usually laid flat, and asa result, the effective limit on the width of a brick is set by the distance which can conveniently be spanned between the thumb and fingers of one hand, normally about 100 mm (4 in). In most cases, the length of a brick is twice its width plus the width of a mortar joint, about 200 mm (8 in) or slightly more.

FACE BRICK ("HOUSE BRICK") SIZES, (ALPHABETICAL ORDER)

STANDARD	METRIC (MM)	IMPERIAL(INCHES)
AUSTRALIA	230*110*76	9.1*4.3*3.0
CHINA	240*155*53	9.4*6.1*2.1
DENMARK	228*108*54	9.4*4.3*2.1
GERMANY	240*115*71	9.4*4.5*2.8
INDIA	228*107*69	9.0*4.2*2.7
JAPAN	210*100*69	8.3*3.9*2.4
ROMANIA	240*115*63	9.4*4.5*2.5
RUSSIA	250*120*65	9.4*4.7*2.6
SOUTH AFRICA	222*120*62	8.7*4.2*2.9
SWEDEN	250*120*62	9.8*4.7*2.4
U.K	215*102.5*65	8.5*4.0*2.6
U.S	194*92*57	7.6*3.6*2.2



Uses : Bricks are a versatile building material, able to participate in a wide variety of applications, including :

- Structural walls, exterior and interior walls.
- Bearing and non-bearing sound proof partitions.
- The fireproofing of structural-steel members in the form of firewalls, party walls, enclosures and fire towers.
- Foundations for stucco.
- Chimneys and fireplaces.
- Porches and terraces.
- Outdoor steps, brick walks and paved floors.
- Swimming pools

Papercrete is a building material that consists of re-pulped paper fiber with Portland cement or clay and/or other soil added. First patented in 1928, it was revived during the 1980s. Although perceived as an environmentally friendly material due to the significant recycled content, this is offset by the presence of cement. The material lacks standardization, and proper use therefore requires care and experience. Eric Patterson and Mike McCain, who have been credited with independently "inventing" papercrete (they called it "padobe" and "fibrous cement"), have both contributed considerably to research into machinery to make it

And ways of using it for building. The paper to be used can come from a variety of sources, including newspaper, junk mail, magazines, books. A mixer is used to pulp the mix. Depending on the type of mixer, the paper may be soaked in water beforehand. As of 2007, papercrete lacks approval from the International Code Council. This limits its range of use within the city limits of most incorporated United States citieswhere building codes apply. It is not used as a load-bearing wall where building codes apply. However, its strength in model structures has been proven, and homes and small commercial buildings are being constructed. There is little or no evidence of its long-term durability. In these small building projects, papercrete is being used as an in-fill wall in conjunction with structural steel beams or other load-bearing elements. Papercrete gets its name from the fact that most formulas use a mixture of water and cement with cellulose fiber.

CHAPTER - 5
EXPERIMENTAL WORK

5.1. COMPONENTS :

1 . PAPER : Paper is a thin sheet material produced by mechanically or chemically processing cellulose fibers derived from wood, raps, grasses or other vegetable sources in water, draining the water through fine mesh leaving the fiber evenly distributed on the surface, followed by pressing and drying. Although paper was originally made in single sheets by hand, almost all is now made on large machines. It is a versatile material with many uses, including printing, painting, graphics, signage, design, packaging, decorating, writing, and cleaning. It may also be used as filter paper, wallpaper, book endpaper, conservation paper, laminated worktops, toilet tissue, or currency and security paper, or in a number of industrial and construction processes.

The tiny small pieces of the paper is used in the mix design for the manufacturing of paper brick.

{ Small Pieces Of Waste Paper }



2. CEMENT : A cement is a binder, a chemical substance used for construction that sets, hardens, and adheres to other materials to bind them together. Cement is seldom used on its own, but rather to bind sand and gravel(aggregate) together. Cement mixed with fine aggregate produces mortar for masonry, or with sand and gravel, produces concrete. concrete is the most widely used material in existence and is behind only water as the planets most consumed resource. cements used in construction are usually inorganic, often lime or calcium silicate based, which can be characterized as hydraulic or the less common non-hydraulic, depending on the ability of the cement to set in the presence of water. Hydraulic cements (e.g, Portland cement) set and become adhesive through a chemical reaction between the dry ingredients and water. The chemical reaction results in mineral hydrates that are not very water-soluble and so are quite durable in water and safe from chemical attack.

{ Ordinary Portland cement }



3 . SAND : Sand is granular material composed of finely divided mineral particles. Sand has various compositions but is defined by its grain size. Sand grains are smaller than gravel and coarser than silt. Sand can also refer to a textural class of soil or soil type; i.e. a soil containing more than 85% sand-sized particles by mass.

{ Sand(fine aggregate) used in the mix design }



4 . PET Plastic Bottles : A plastic bottle is a bottle constructed from high-density or low-density plastic. Plastic bottles are typically used to store liquids such as water, soft drinks, motor oil, cooking oil, medicine, shampoo, milk, and ink. The size ranges from very small bottles to large carboys. Consumer blow molded containers often have integral handles or are shaped to facilitate grasping. in this project we use small pieces of plastic. So that it can be mixed properly.

{ Plastic bottle crushed into small pieces to use in design mix }



5 . WATER : Water is an inorganic compound with the chemical formula H2O. it is a transparent, tasteless, odorless, and nearly colorless chemical substance, and it is the main constituent of earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent). It is vital for all known forms of life, despite not providing food, energy or organic micronutrients. Its chemical formula, H2O. indicates that each of its molecule contains one oxygen and wo hydrogen atoms, connected by covalent bonds. The hydrogen atoms are attached to the oxygen atom at an angle of 104.45 degrees. "Water" is also the name of liquid state of H2O at standard temperature and pressure.

{ Water should be added for the better mixing of components to form the papercrete }



5.2. PROCEDURE :

In this section we will explain how the paper bricks can be produced using waste paper. Portland cement, and PET(polyethylene terephthalate) bottles. The detailed step-by-step method of paper brick making is discussed below**.**

1. First of all we collect the components used for the manufacturing of paper brick. i.e. (waste paper, cement, sand , waste PET bottles, water)
2. Waste papers are collected and broken down into small pieces.
3. Then the pieces of paper are soaked In water for 24 hours and that will grinded (manually or by using machines) into smaller lumps for easy mixing with sand and cement.



4. These lumps are taken out from the soaked water and kept under the sunlight for wetting process(equally 24hrs).



5. Now, the paper lumps are get wetted. So, these paper lumps are ready to use.
6. The lumps which we have are now converted into granules with a mixture of Portland cement slurry, sand and paper with the proportions of 1:1:5.
7. Fly ash is also used in the mix design for the manufacturing of brick. It reduces the actual weight of brick. and makes it a light weight brick.(tested experimentally by absorption test)
8. Portable water is used as per proportion ratio for the design mix.
9. These mixture is mixed properly as per ratios.



10. Now, these formed solution is taken into 19*4*3 dimensioned iron brick mould.



11. These papercrete mould is subjected to vibrator machine process to compact the solution properly from which it gets exact shape and size.





12. Extra wastage is removed from the block which undergo after the machine process.
13. Now, the blocks are reserved on a plane surface to get exact copy of a mould.
14. These bricks should be wet under sunlight for 24 hours.



15. Now, the bricks are undergo curing process in curing ponds.





5.3. EXPERIMENTAL TESTS :

In this project we are going to perform 3 Lab Experiments Tests on our paper brick samples, after the completion of their curing time period. That are as follows :

1. WATER ABSORPTION TEST :

Water absorption test on bricks are conducted to determine durability property of bricks such as degree of burning, quality and behavior of bricks increase with increase in pores. Brick for external use must be capable of preventing rain from passing through them to the inside of walls of reasonable thickness. A good brick should absorb water maximum $1/7^{th}$ of the weight of the brick. The degree of compactness of bricks can obtained by water absorption test, as water is absorbed by pores in bricks. The water absorption by bricks increase with increase in pores. Water absorption shall not be more than 20%. Water absorption test is required to check whether the bricks are suitable for water logged areas or not. As per standards the bricks should not absorb water more than 20% of its original weight. The water absorption obtained for conventional brick is 16.66%. All the papercrete bricks have higher water absorption than the conventional brick.

$$\textbf{Water absorption} = [(W_2 - W_1)/W_1] * 100$$

W_1 = dry weight of paper brick.

W_2 = wet weight of paper brick.







2. COMPRESSION TEST :

In order to determine the load carrying capacity under compression compressive strength testing was conducted on bricks. Using the compression testing machine, this experiment was carried out.

As per IS 3495:1992, 3.5 N/mm2 is standard compressive strength for conventional brick. Compression test is that the main and vital test for bricks. This test was dispensed by a Compression Testing Machine (CTM). Instrument is set up for applying load. Brick is placed between the plates of testing machine. Load is axially applied at uniform rate of 14 N/mm2 per minute till failure occurs and maximum load at failure is noted. Compressive stress is calculated using equation F= P/A N/mm2 Where, P = Load in N A = Area of the specimen in Sq.mm.

COMPRESSIVE STRENGTH (F) = P/A (N/SQ.MM)

P = Peak load (N)

A = Area (Sq.mm)











3. FIRE RESISTANCE TEST :

The sample was made to be examined for fire resistance, by exposing the sample to fire, during the process it was found that the sample did not caught fire. The temperature was also found to be not transferred to the other side of the sample. Sample was made to undergo fire resistance test in an electrical oven at a temperature of 150 C . After 4 hours of heating it was found that physical state of brick was good and sound without any deformation and failure to fire test.

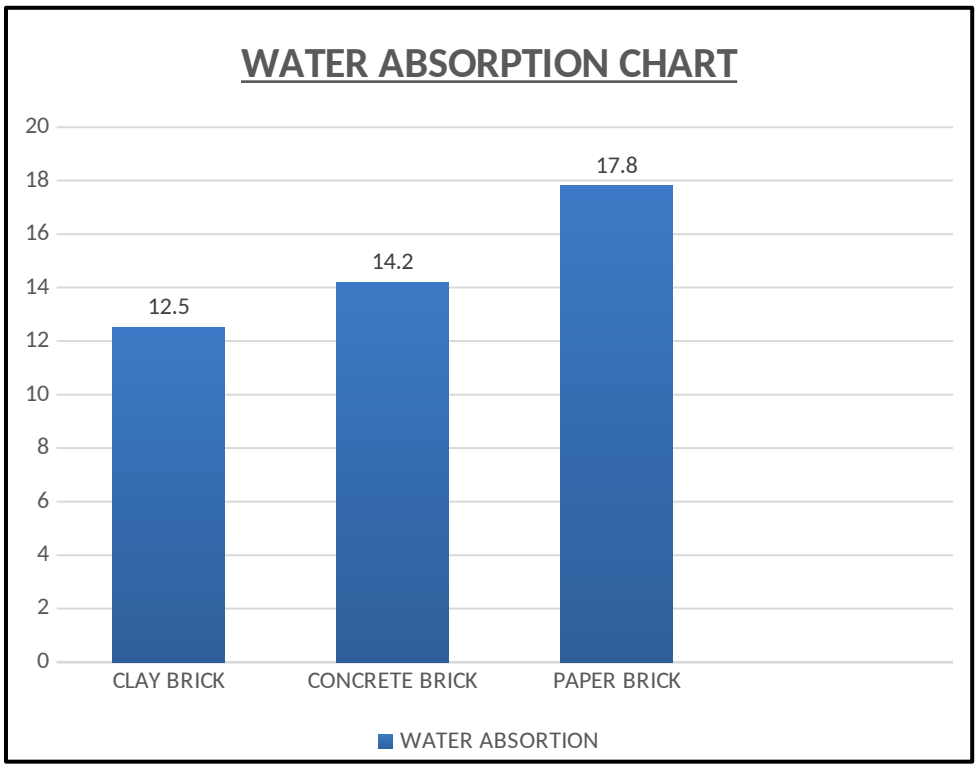
CHAPTER - 6
RESULTS

6.1. WATER ABSORPTION TEST :

1. Tabular representation of the water absorption values of different bricks

TYPE OF BRICK	WATER ABSORPTION VALUE (%)
CLAY BRICK	12.5 %
CONCRETE BRICK	14.2 %
PAPER BRICK	17.8 %

2. Graphical representation of the water absorption values of different bricks

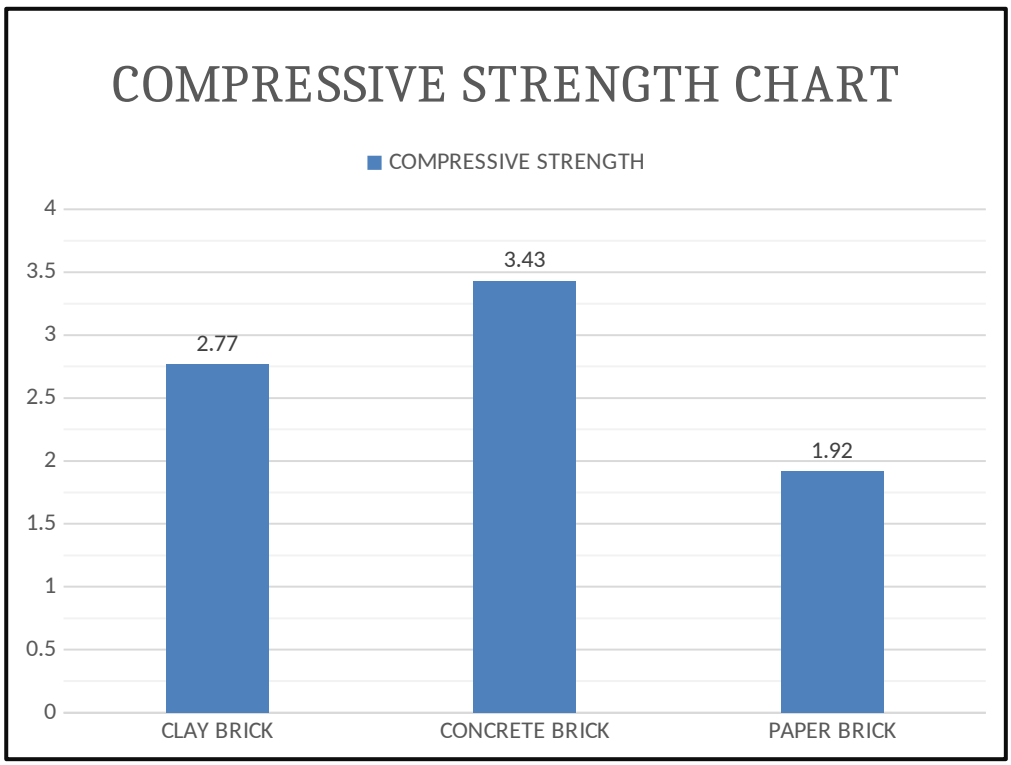


6.2. COMPRESSION TEST :

1. Tabular representation of the compressive strength values of different bricks

TYPE OF BRICK	COMPRESSION STRENGTH (MPa)
CLAY BRICK	2.77 Mpa
CONCRETE BRICK	3.43 MPa
PAPER BRICK	1.9 MPa

2. Graphical representation of the compressive strength values of different bricks



CHAPTER - 7
CONCLUSION

- Compared to conventional brick, the paper brick is light in weight.
- Paper bricks are appropriate for non-load bearing walls. These bricks don't seem to be appropriate for water logging and external walls. It can be utilized in inner partition walls.
- Due to less weight of the paper bricks, the dead load of the building will be reduced.
- Due to lesser weight and more flexibility, these bricks are potentially ideal material for earthquake prone regions.
- These bricks does not expand nor shrinks hence, sheets of glass or glass block can be fixed in and trimmed with papercrete.
- Since, paper brick mainly consists of waste material, it will reduce the landfills and pollution. Hence, the overall cost is very low as compared to conventional brick. It has been seen that by using paper bricks in a building construction, the total cost was reduced from 20% to 50%.
- It is apparent from the above researches that various types of waste materials from the different industries have been used in different proportions and different methods are adopted to produce bricks.
- Based on the literature review, the researchers have revealed that many successful attempts are made to incorporate waste materials in the manufacturing of bricks. They have positively influenced and enhanced the performance in terms of making them environment friendly and manufacturing economical bricks leading to the design of green building.
- Certain bricks are made without firing which is an advantage over other method of manufacturing of bricks in terms of low embodied energy material.
- It also offers a solution to the problem of waste disposal as well as eco-friendly environment in construction industry.
- The use of eco-friendly bricks made-up of waste materials is still very limited and there is a scope of further research in manufacturing reinforced bricks using waste materials. Further research and development is needed to promote wide range of production of

eco-friendly low cost bricks from waste materials in the absence of relevant standards, which have semi acceptance by the industry and the public.

- Based on the experimental investigation the subsequent effects were determined. The superior (%) Of cement substitute in paper brick and compressive energy of the bricks have studied. Paper bricks extra low-priced the time required to prepare bricks much less in comparison to normal bricks.
- Brick is one of the major components used for construction and it has a significant impact on the construction costs of dwellings of urban poor.
- Intention of this work is to come up with the cost effective alternative to conventional building material.
- Experimental results shows that there is a very vast scope for developing cost effective alternative building materials.
- Lack of official guidelines and standard procedure describing the method of producing papercrete.
- Papercrete is far lighter a material than concrete or wood and has good water absorption and insulation capacity, less shrinkage and is environment friendly, has high strength to weight ratio.
- Although this area requires significant amount of research, but papercrete can ideally be used for reduction in dead weight of structural elements in cases where there are no special loads acting on the element. Also, about 55% of paper waste around the globe still does not get recycled and is directly disposed of and a considerable portion of it can be used for manufacturing papercrete. Future research on improving its tensile strength can open up the possibilities of papercrete using it as a construction worldwide.
- Sustainable development brings out stability in the requirement of the environment. It makes the resources available for use for the future generations. Sustainable development is an amazing way to conserve the resources provided by nature.
- Because of the cheapest construction by this papercrete poor people will construct easily.
- Good choice to utilize waste paper.
- The construction done by the papercrete brick are very light in weight. So in earth quake regions these are very helpful by reducing the life losses and the property losses.
- Papercrete bricks satisfy the basic characteristics of conventional bricks study.

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